

Agenda

→ Train Test split

→ Forecasting Method

↳ Simple Method

↳ Smoothing Method

→ forecasting (Predict Sales)

↳ $MAPE < 5\%$

$$MAPE = \frac{1}{n} \sum \left| \frac{y^{(i)} - \hat{y}^{(i)}}{y^{(i)}} \right| \times 100\%$$

↳ Relative deviation from $y^{(i)}$

MAPE

MAE = 1

←

$$\begin{aligned} y^{(i)} &= 10 \\ \hat{y}^{(i)} &= 9 \end{aligned}$$

MAPE

MAE = 1

←

$$\begin{aligned} y^{(i)} &= 1000 \\ \hat{y}^{(i)} &= 999 \end{aligned}$$

$y^{(i)}$ Sales = 350
 $\hat{y}^{(i)}$ = Predict = 315 $\rightarrow$ Underforecasting

$$MAPE = \frac{350 - 315}{350} = 10\%$$

Overforecasting $y^{(i)} = 350$
 $\hat{y}^{(i)} = 385 \leftarrow$

$$MAPE = \frac{385 - 350}{350} = 10\%$$

→ Train test split 80-20%. → Cannot have missing values in time series

Date	y
2001-01-01	
2001-02-01	
2001-03-01	
2001-04-01	
2001-05-01	
2001-06-01	

Train		80%.
Test		20%.

80% - 20%

$t-100 \quad t-99 \quad \dots \quad t$

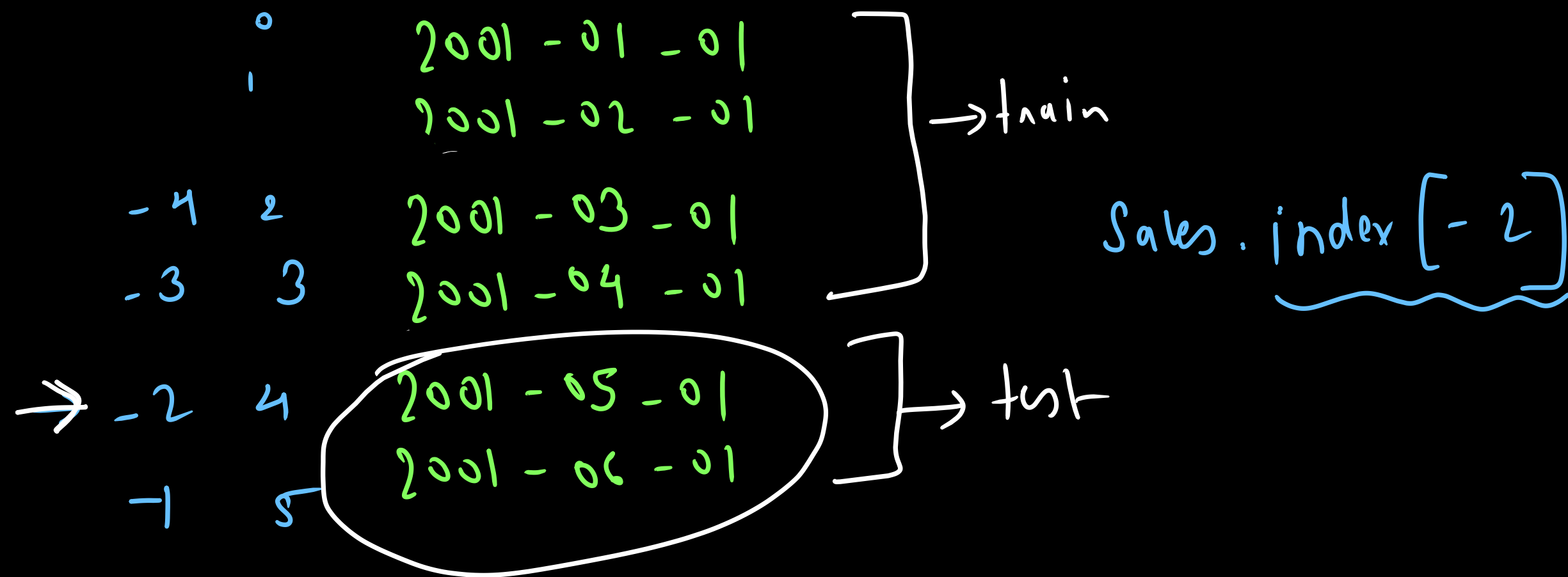
Date	y
2001-01-01	
2001-02-01	
2001-03-01	
2001-04-01	
2001-05-01	
2001-06-01	

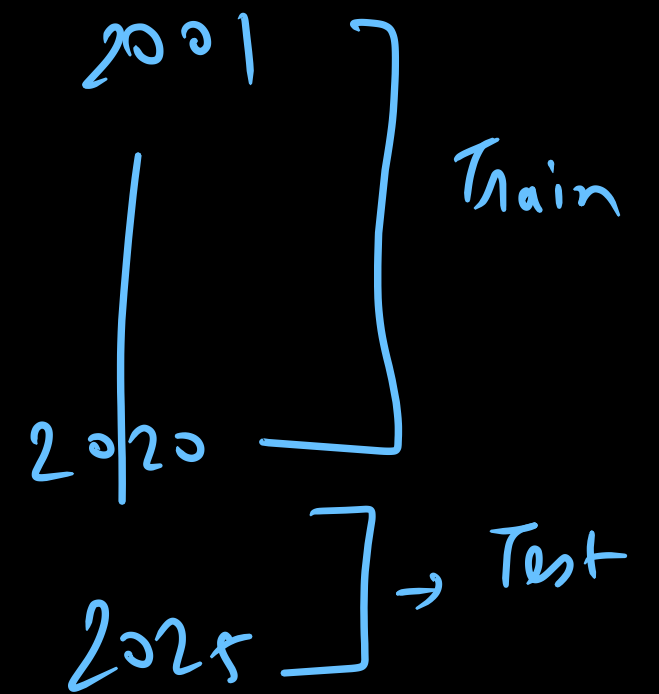
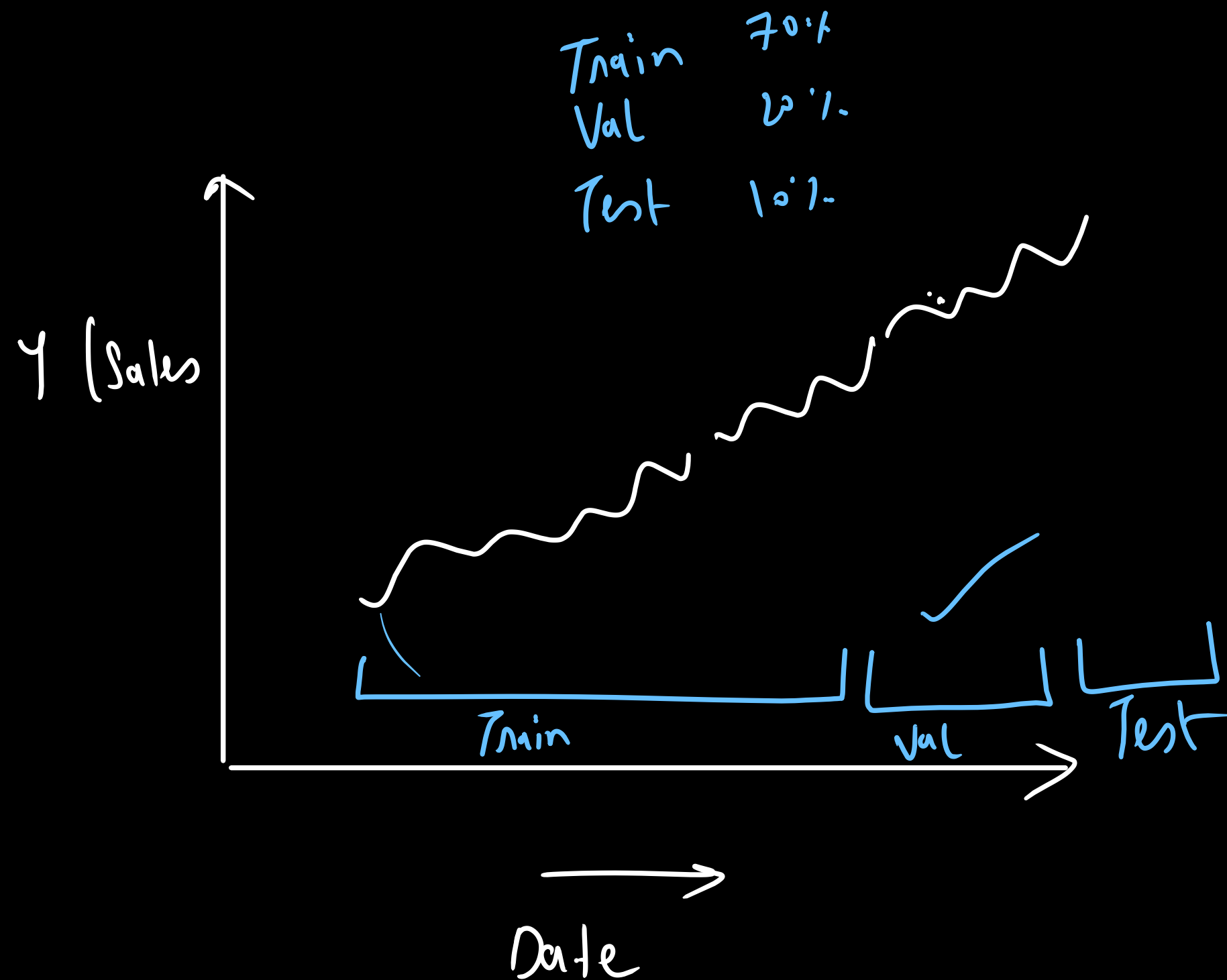
Train

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Test

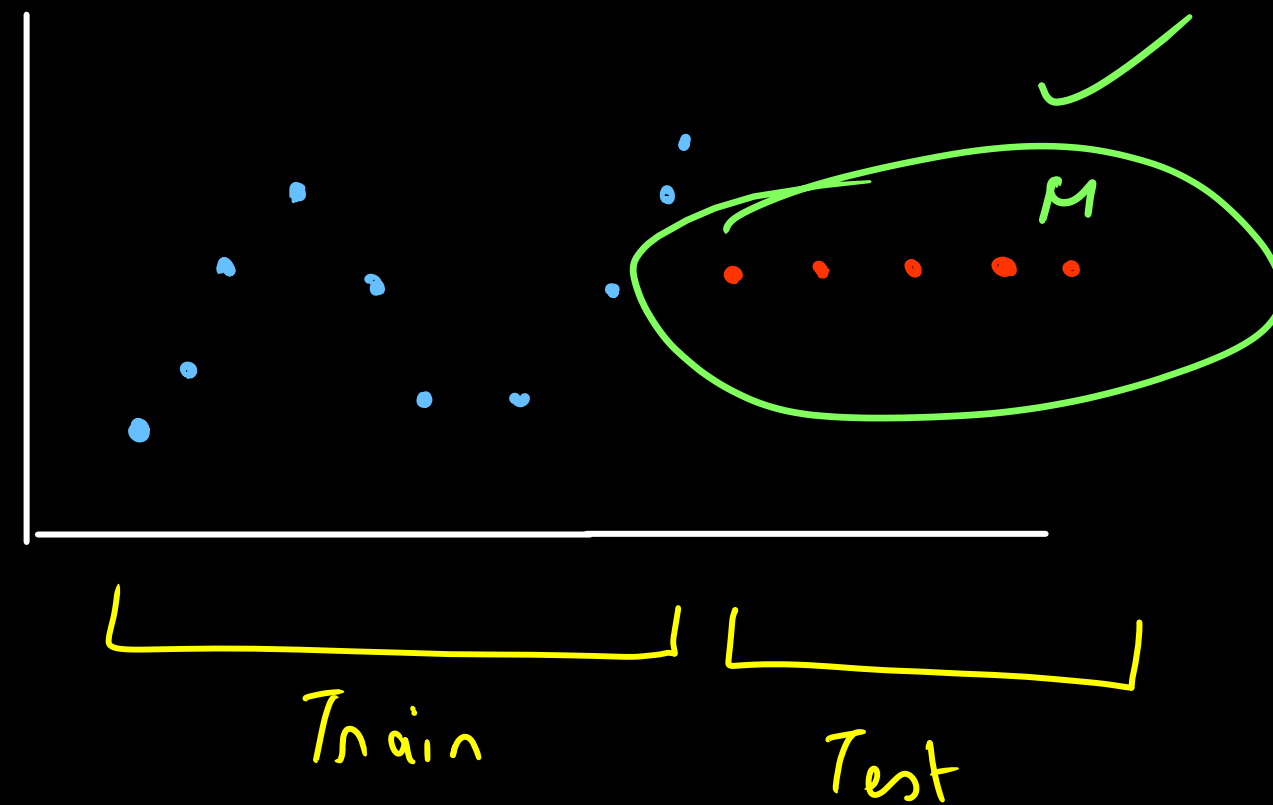
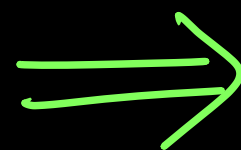
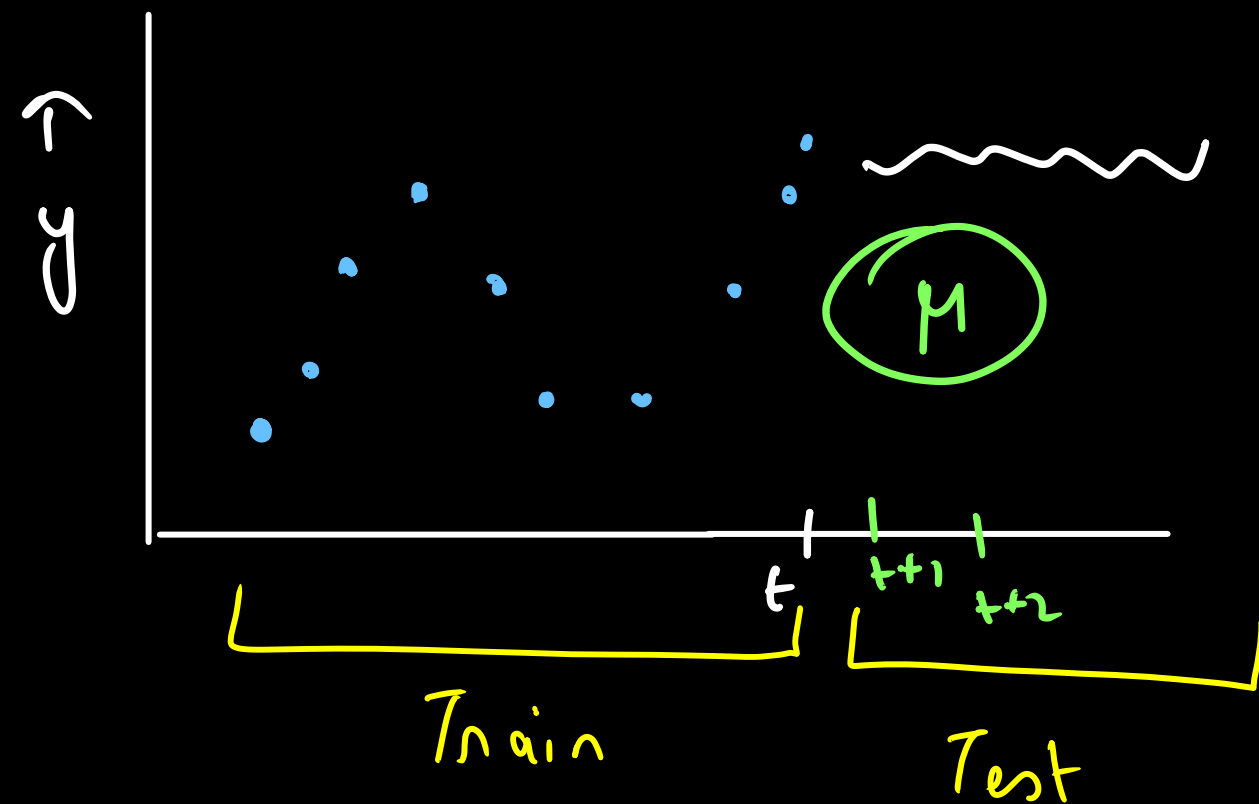
2001-05-01	
2001-06-01	





→ Simple Forecasting Method

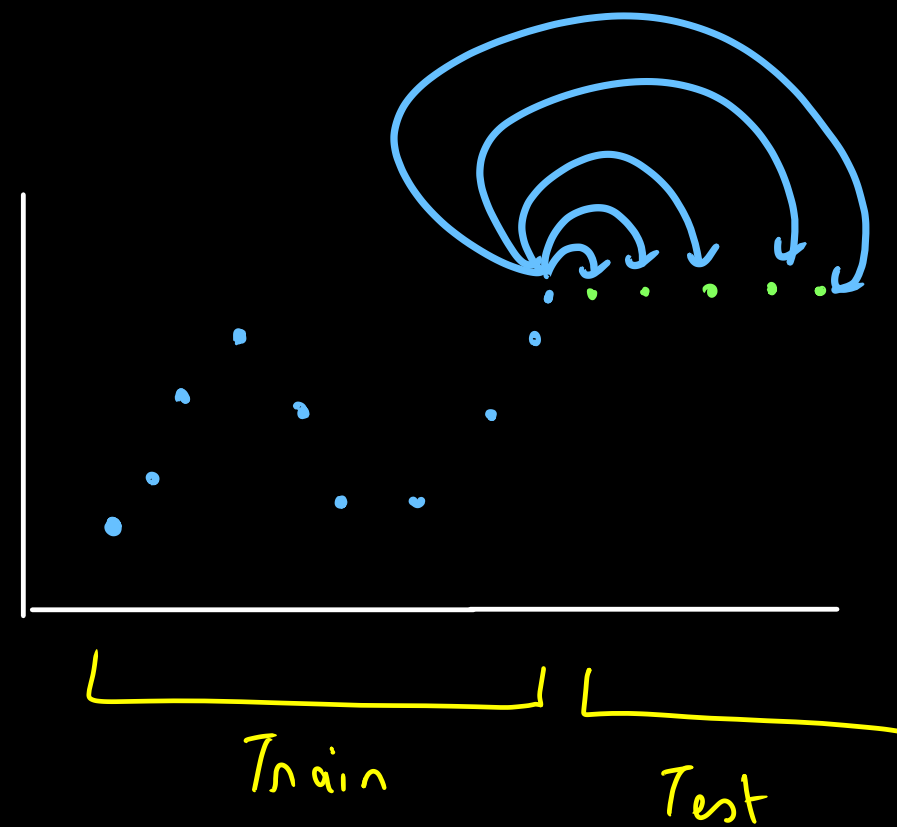
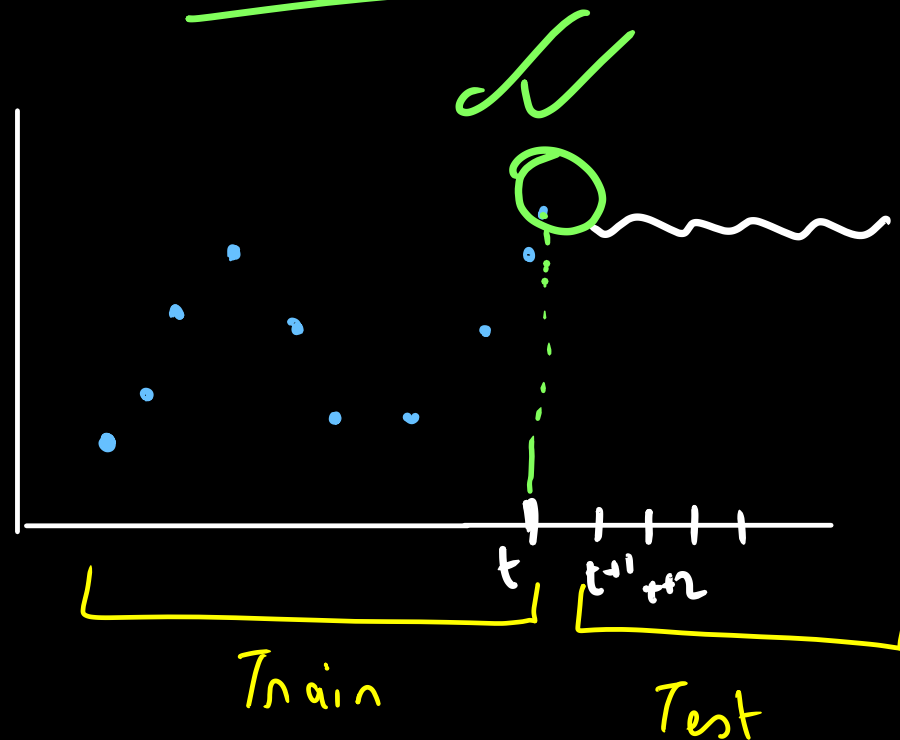
→ Mean forecasting



Date →
 μ = Mean of training data

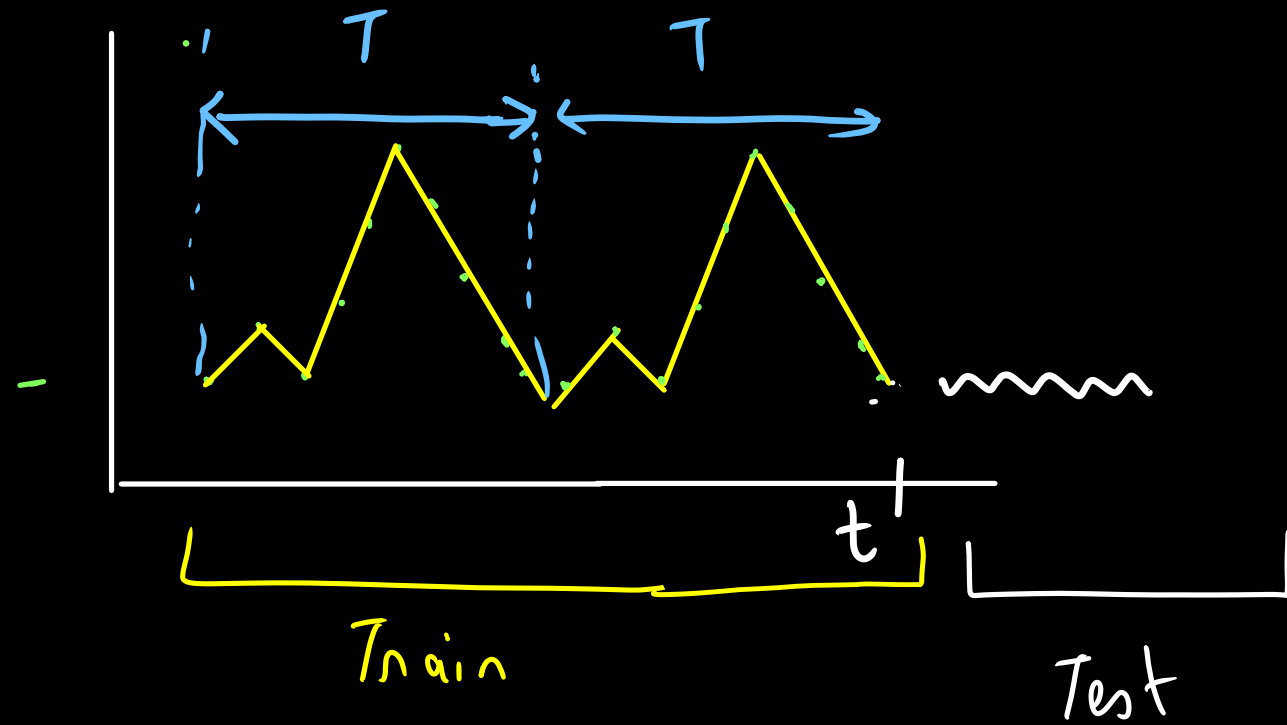
$$\begin{cases} \hat{y}_{t+1} = \mu \\ \hat{y}_{t+2} = \mu \\ \vdots \\ \hat{y}_{t+h} = \mu \end{cases}$$

→ Naive Approach

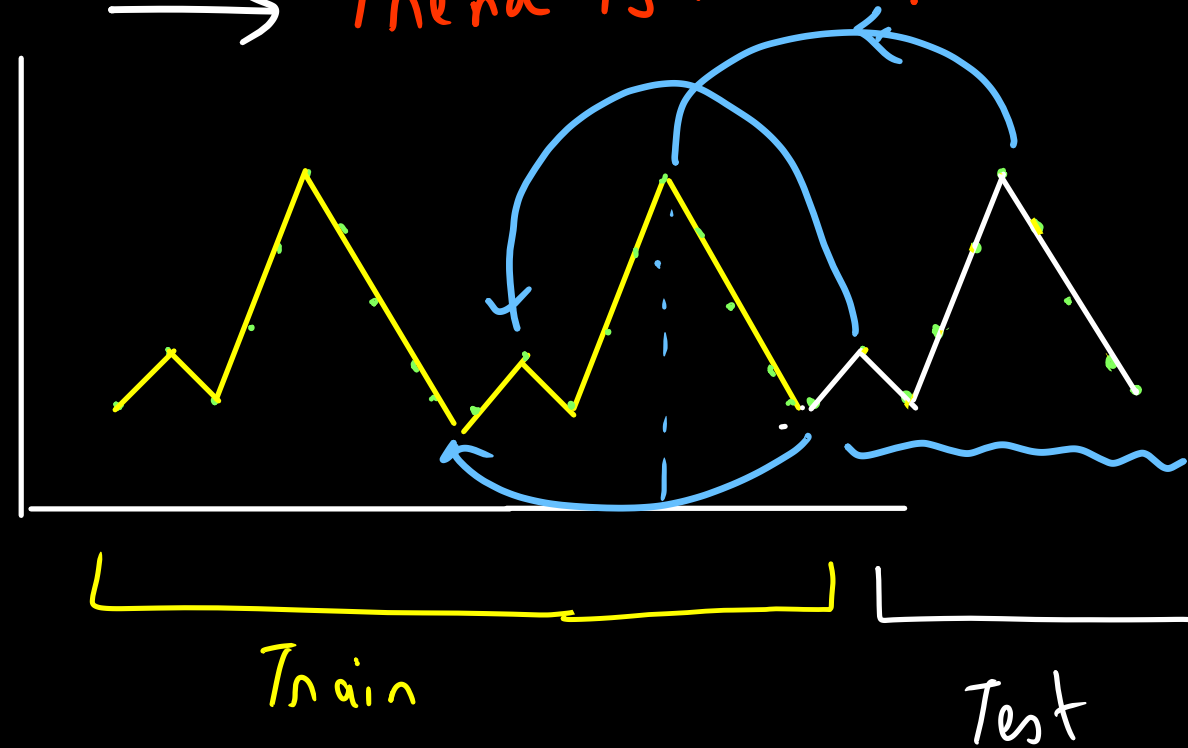


$$\begin{cases} \hat{y}_{t+1} = y_t \\ \hat{y}_{t+2} = y_t \\ \hat{y}_{t+3} = y_t \end{cases}$$

→ Seasonal Naive Forecast



→ Seasonality is present
 → Trend is not captured



$$\hat{y}_{t+h} = y_{t-T+h}$$

$$\left\{ \begin{array}{l} \hat{y}_{t+1} = y_{t-T+1} \\ \hat{y}_{t+2} = y_{t-T+2} \end{array} \right.$$

$T = \text{Seasonality}$

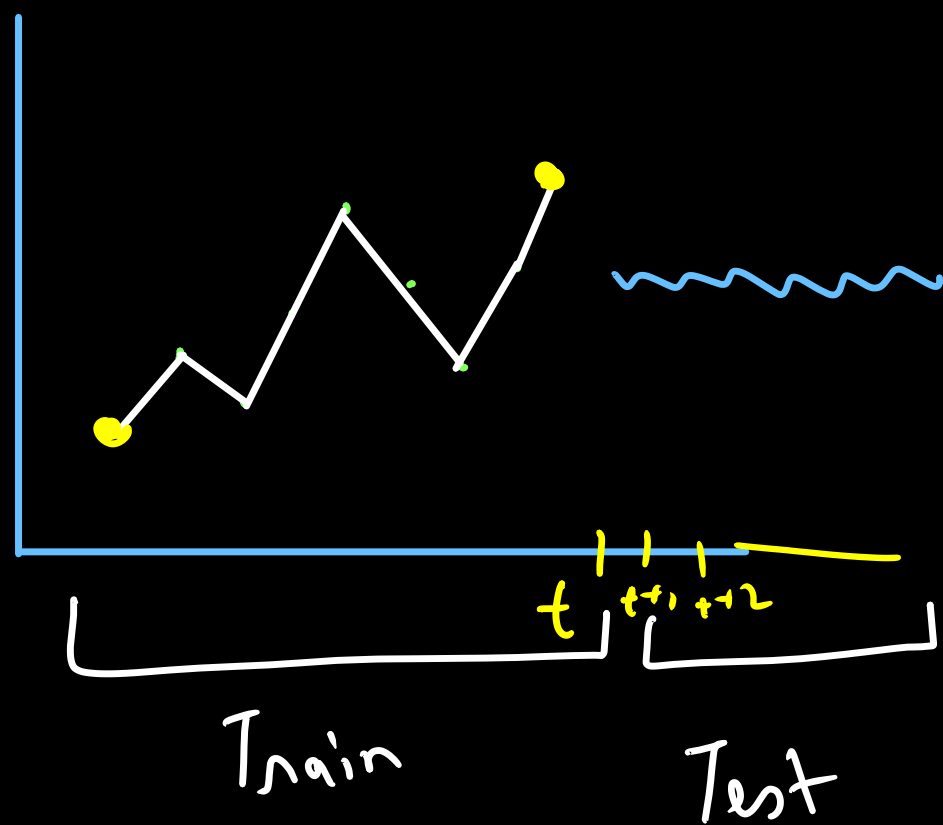
✓ $T \varepsilon$

① statsmodel

② Auto correlation

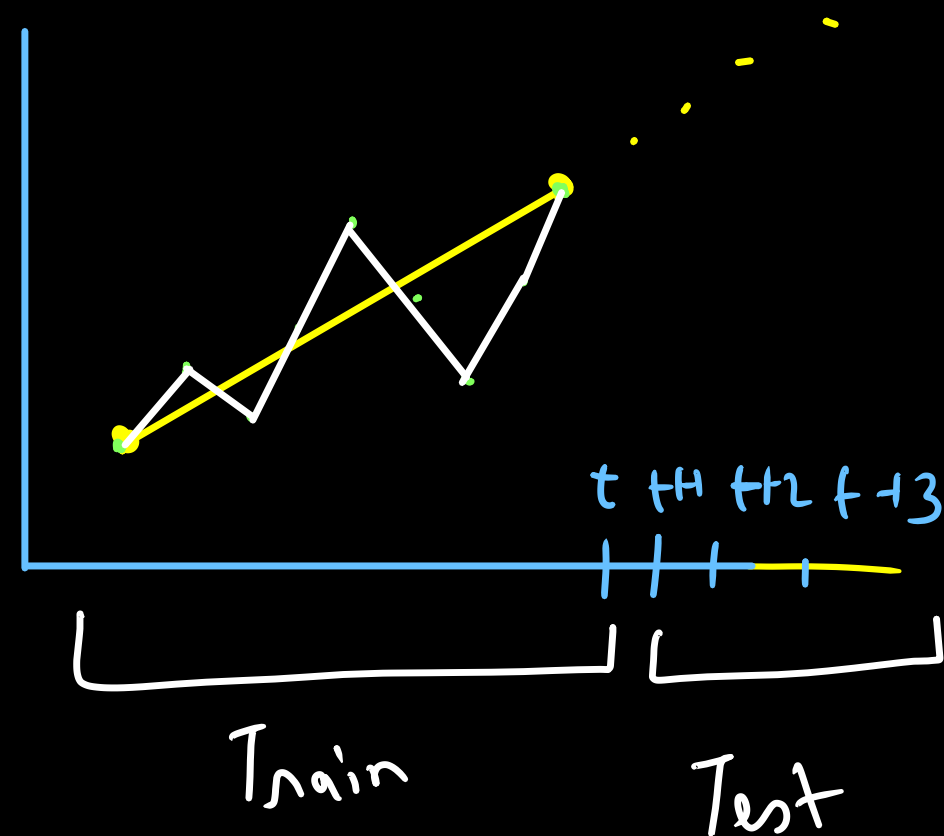
→ Break until 22:34 PM

→ Drift Method



$$\text{Slope} = \frac{y_t - y_0}{t}$$

→ Trend



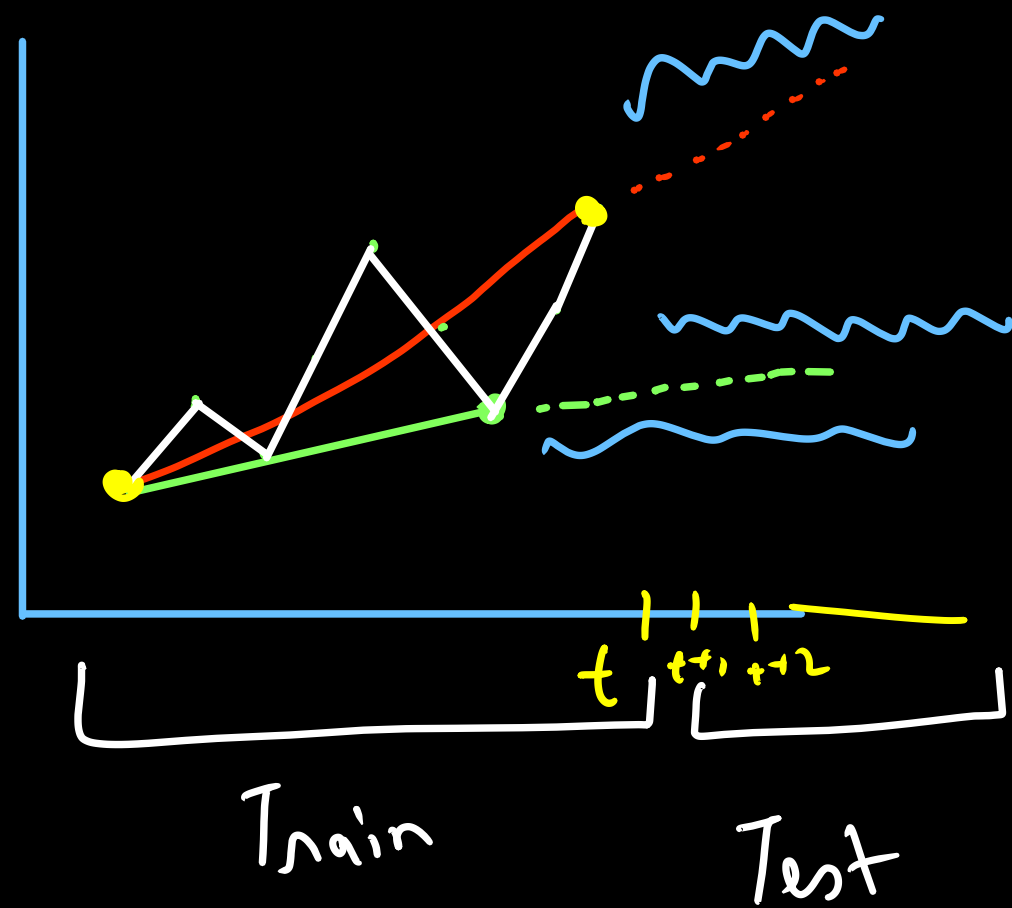
$$\hat{y}_{t+1} = y_t + 1 \cdot \text{Slope}$$

$$\hat{y}_{t+2} = y_t + 2 \cdot \text{Slope}$$

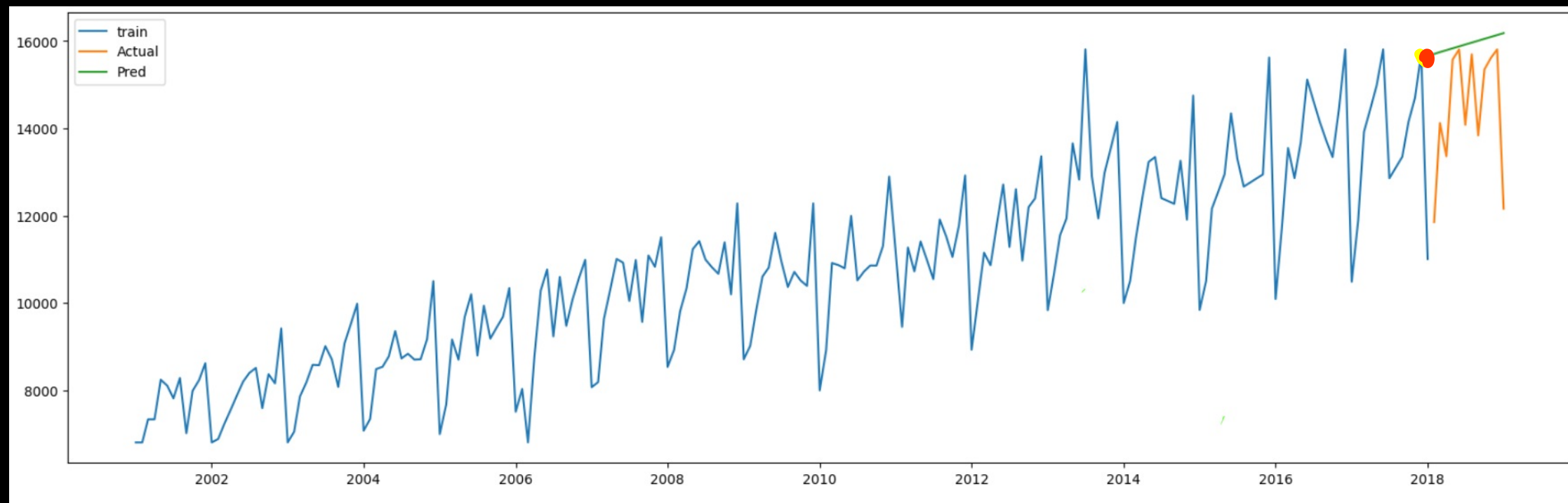
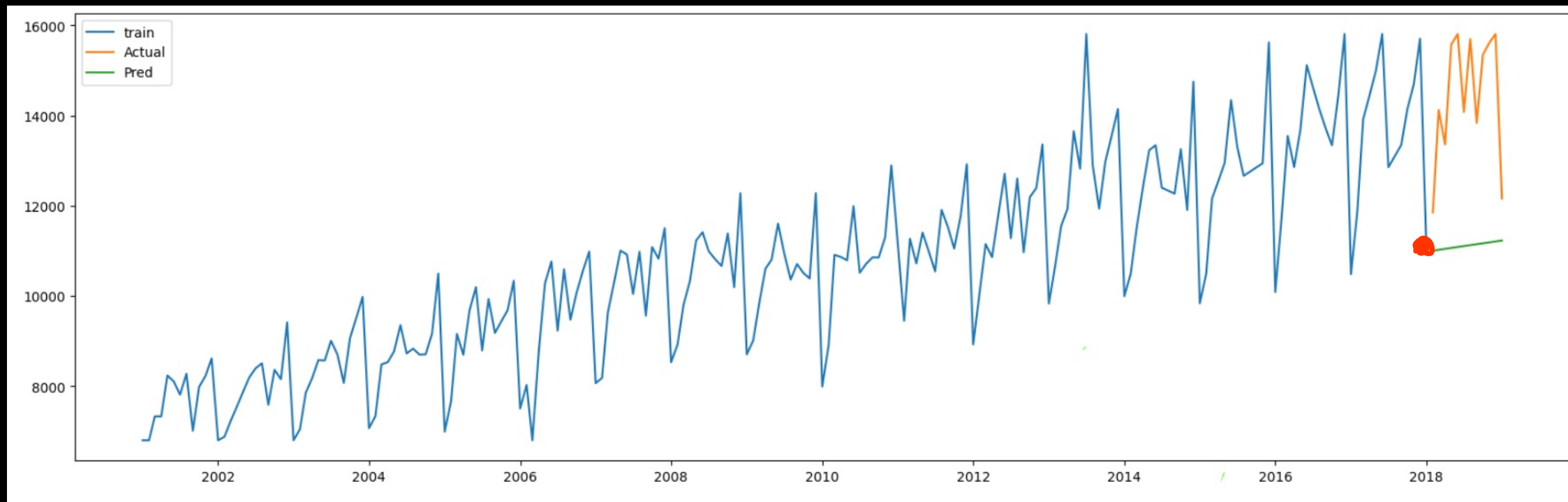
$$\hat{y}_{t+3} = y_t + 3 \cdot \text{Slope}$$

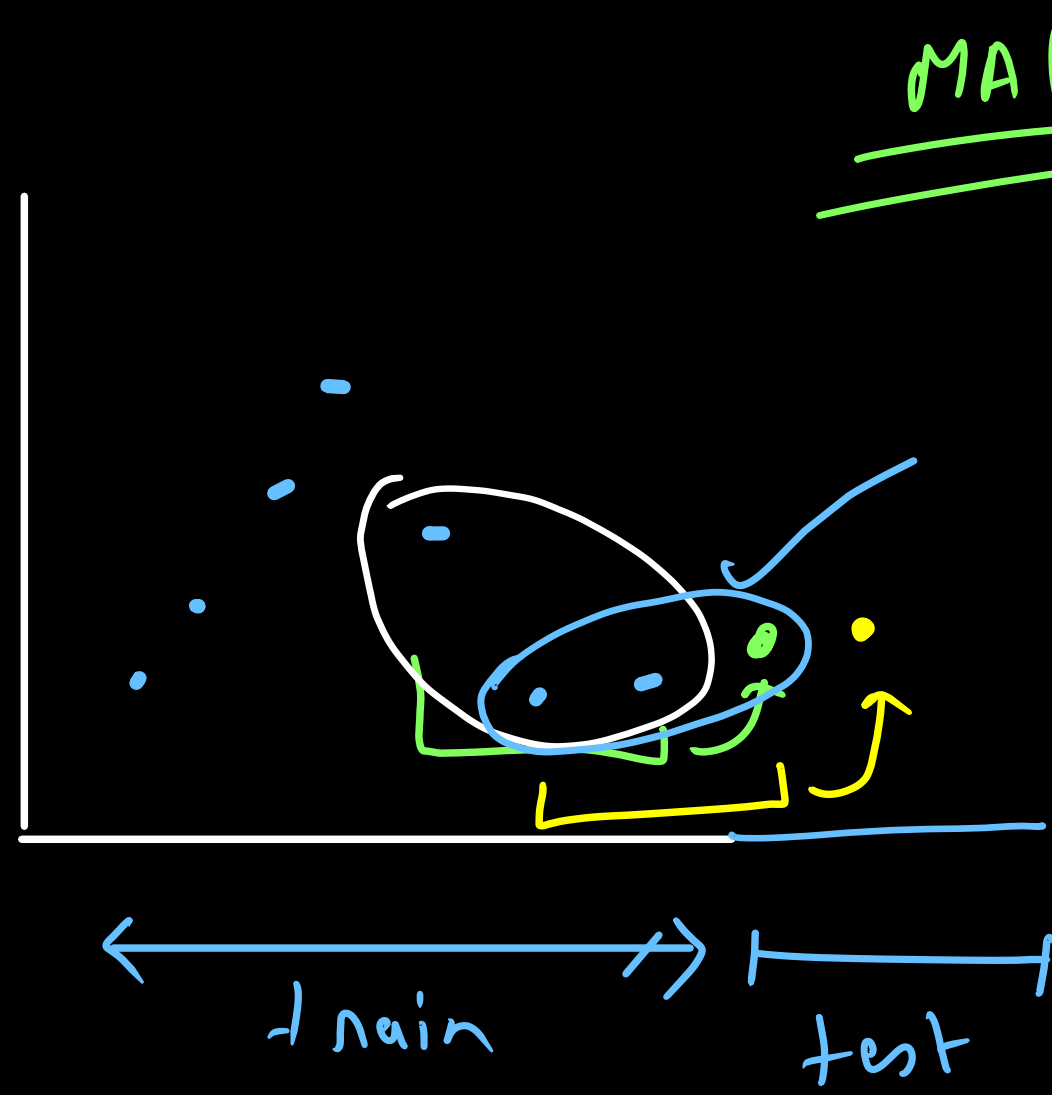
$$\hat{y}_{t+h} = y_t + h \cdot \text{Slope}$$

✓ → Highly sensitive to last data point



$$\frac{y_f - y_0}{\text{time}}$$





Max weigh

↓ ✓

✓

Min weigh

↓ ✓

$$\hat{y}_t = \frac{y_{t-1} + y_{t-2} + y_{t-3}}{3}$$

$$\hat{y}_t = \frac{w_1 y_{t-1} + w_2 y_{t-2} + \dots + w_m y_{t-m}}{m}$$

Max weightage
↓
↓ Min weigh

$$= \frac{1}{m} \sum_{i=t-m}^{t-1} y_i$$

→ Exponential Smoothing

$$\hat{y}_{t+1} = L_t$$

$$L_t = \alpha y_t + (1-\alpha) \cdot L_{t-1}$$

$$= \alpha y_t + (1-\alpha) [\alpha y_{t-1} + (1-\alpha) L_{t-2}]$$

α = Smoothing constant

$$= \alpha y_t + \alpha(1-\alpha) y_{t-1} + (1-\alpha)^2 L_{t-2}$$

$$\hat{y}_{t+1} = L_t$$

$$L_t = \alpha y_t + \alpha(1-\alpha) y_{t-1} + (1-\alpha)^2 \alpha y_{t-2} \dots$$

High $\downarrow$

less $\downarrow$

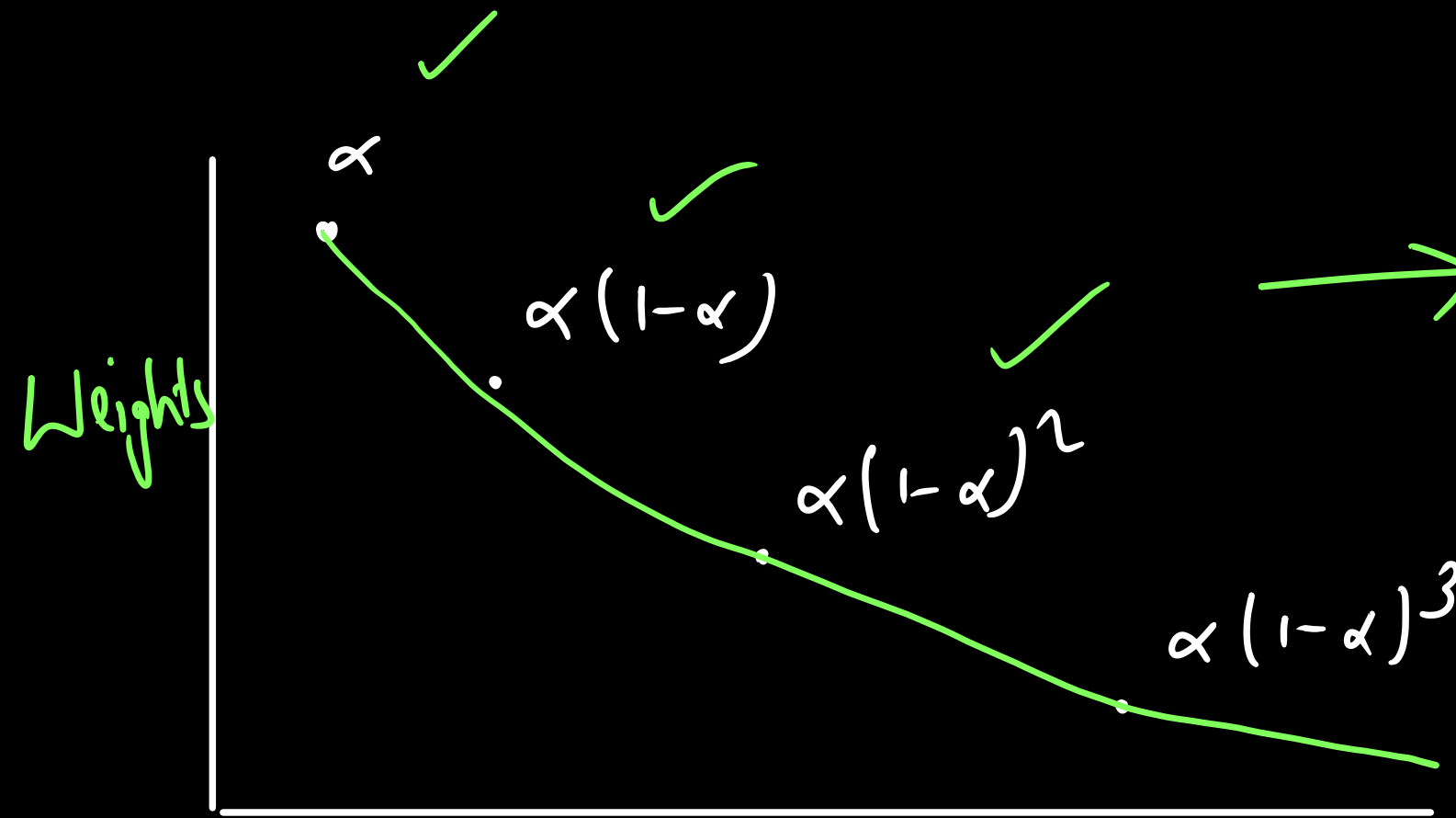
lessen $\downarrow$

$$= 0.2 y_t + 0.2 \times 0.8 y_{t-1} + (1-0.2)^2 0.2 y_{t-2} \quad \alpha = 0.2$$

$$= 0.2 y_t + 0.16 y_{t-1} + 0.128 y_{t-2} \dots$$

$\downarrow$
Most recent

$\downarrow$



→ Exponential
decay